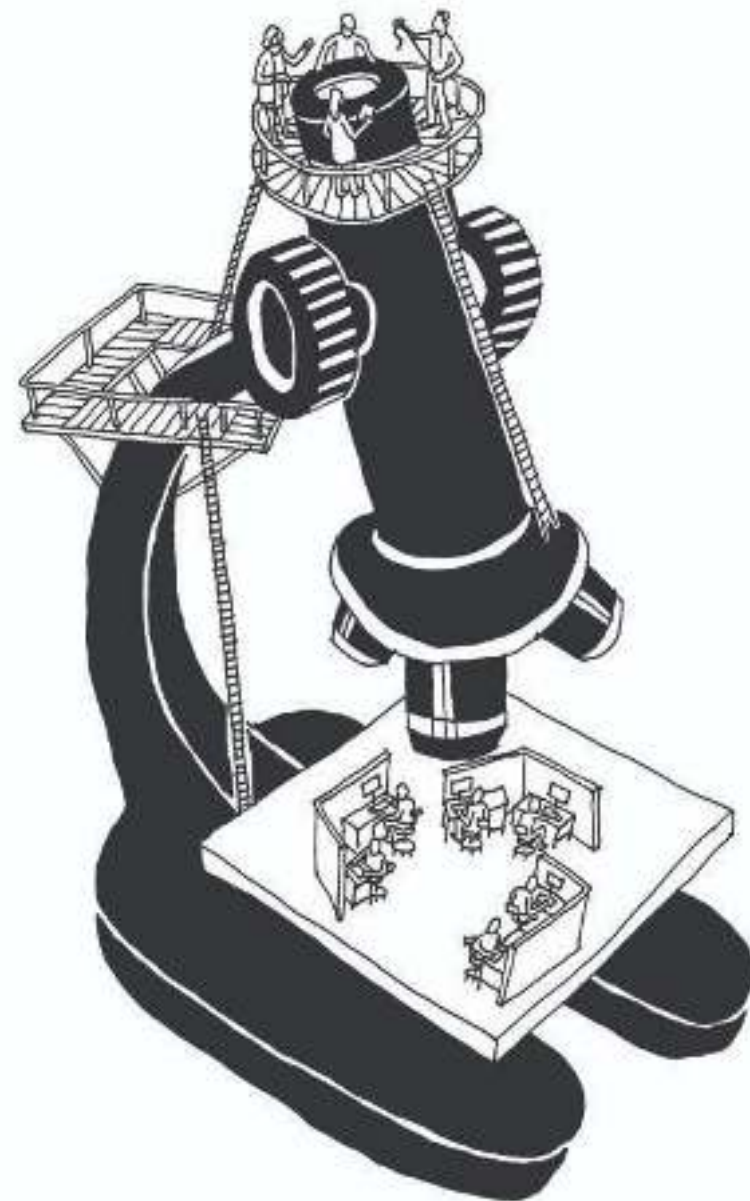


ECS 260: Software Engineering

Lecture 2: Research Methods and Questions

(slides by Eastbrook and Vasilescu)



Planning Checklist

- ✓ Pick a topic
- Identify the research question(s)
- Check the literature
- Identify your philosophical stance
- Identify appropriate theories
- Choose the method(s)
- Design the study
 - Unit of analysis?
 - Target population?
 - Sampling technique?
 - Data collection techniques?
 - Metrics for key variables?
 - Handle confounding factors
- Critically appraise the design for threats to validity
- Get IRB approval
 - Informed consent?
 - Benefits outweigh risks?
- Recruit subjects / field sites
- Conduct the study
- Analyze the data
- Write up the results and publish them
- Iterate

Choosing a Research Topic

- *Can* it be studied?

vs

- *Should* it be studied?
 - Does it add to the body of knowledge?
 - Who else besides you would care about results?

Identify the Research Questions

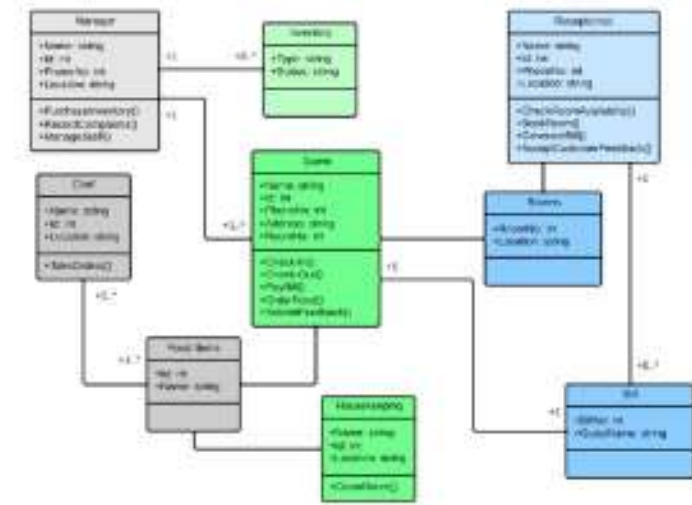
- Where should the questions come from?
 - Research goals
 - Intuition
 - Social Impact
 - Advisor?
- What makes an RQ good?
 - Precise enough
 - Tests a theory
 - No one else has asked in given context

Scenarios - Jane



- Topic: Effectiveness of a novel fisheye-view file navigator (compact format, less scrolling)
- Intuition: fisheye-view file navigator is more efficient for file navigation
- Critics: the more compact information is difficult to read; developers will not adopt it over the traditional file navigator.
- How can Jane find evidence that supports or refutes her intuition?

Scenarios - Joe



- Topic: understanding how developers in industry use (or not) UML diagrams during software design
- Research goals:
 - Explore how widely UML diagrams are used
 - Explore how UML diagrams are used as collaborative shared artefacts during design
- What strategy can Joe follow?

RQs – First Attempt

- Jane: “Is a fisheye-view file navigator more efficient than the traditional view for file navigation?”
- Joe: “How widely are UML diagrams used as collaborative shared artifacts during design?”
- **Thoughts?**

RQs – First Attempt

- Both questions are vague, because they make assumptions about:
 - the phenomena to be studied
 - the kinds of situation in which they occur

RECALL!

What type of question are you asking?

- Existence:
 - Does X exist?
- Description & Classification
 - What is X like?
 - What are its properties?
 - How can it be categorized?
 - How can we measure it?
 - What are its components?
- Descriptive-Comparative
 - How does X differ from Y?
- Frequency and Distribution
 - How often does X occur?
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Better RQs

If Little Prior Knowledge

- Existence questions:
 - “Is file navigation something that (certain types of programmers) actually do?”
 - “Is efficiency actually a problem in file navigation?”
 - “Do collaborative shared artifacts actually exist?”
- Description and Classification questions
 - “How can we measure efficiency for file navigation?”
 - “What are all the types of collaborative shared artifacts?”
- Descriptive-Comparative:
 - “How do fisheye views differ from conventional views?”
 - “How do UML diagrams differ from other representations of design information?”

Better RQs

If Some Understanding of the Phenomena

- Frequency and distribution questions
 - “How many distinct UML diagrams are created in large software companies?”
- Descriptive-Process questions
 - “How do programmers navigate files with existing tools?”
- Relationship questions
 - “Does efficiency in file navigation correlate with the programmer’s familiarity with the environment?”
 - “Do managers’ claims about how often they use UML correlate with the actual use of UML?”

- Causality questions:
 - “Do fisheye-views cause an improvement in efficiency for file navigation?”
- Causality-Comparative questions:
 - “Do fisheye-views cause programmers to be more efficient at file navigation than conventional views?”
- Causality-Comparative Interaction questions:
 - “Do fisheye-views cause programmers to be more efficient at file navigation than conventional views when programmers are distracted?”
- Design questions:
 - “What is an effective way for teams to represent design knowledge to improve coordination?”

Literature Review

- Helps you choose a research topic:
 - Determine if the topic is worth studying
 - Limit the scope
- Ask yourself: How does this project contribute to the literature?
 - Addresses a new topic
 - Uses new data collection method
 - Extends the discussion
 - Replicates a study in a new situation
 - Refines / extends a theory

Purpose of Lit Review

- Share with reader results from related studies
- Relate your study to literature, filling in gaps
- Framework for establishing the importance of your study
- Benchmark for comparing results
- **Provide direction for your research questions and hypotheses**
 - E.g., describe the theory that will be used

Forms of Lit Review

- Integrate what others have done and said
- Criticize prior work
- Build bridges between related topics
- Identify the central issues in a field

How to Structure Lit Review

- Hook
- Coherent presentation of what has been done
- The gap, the need for exactly what you did

Easterbrook et al Chapter

- Selecting Methods -

- Controlled Experiments (Quasi-experiments):
 - determine precisely how variables are related
 - or whether a cause–effect relationship exists
- Case Studies (Exploratory/Confirmatory)
 - offer in-depth understanding of how and why certain phenomena occur
- Survey Research
 - identify the characteristics of a broad population

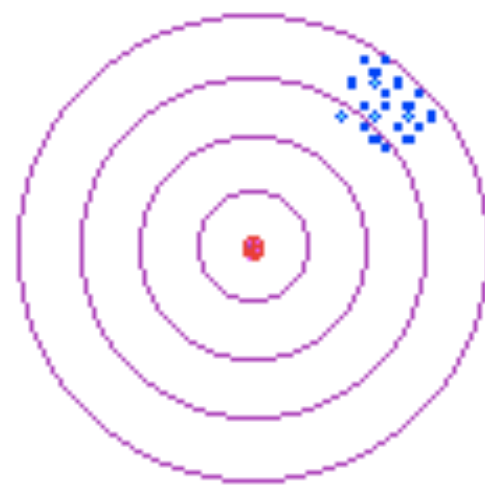
Easterbrook et al Chapter

- Selecting Methods -

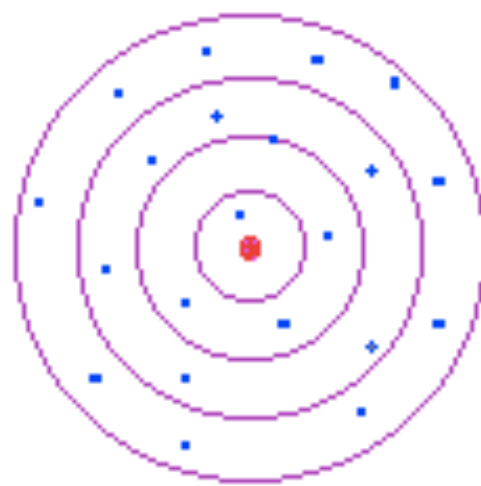
- Ethnographies
 - study a community of people to understand how the members of that community make sense of their social interactions
- Action Research
 - attempt to solve a real-world problem while simultaneously studying the experience of solving the problem

Validity vs. Reliability

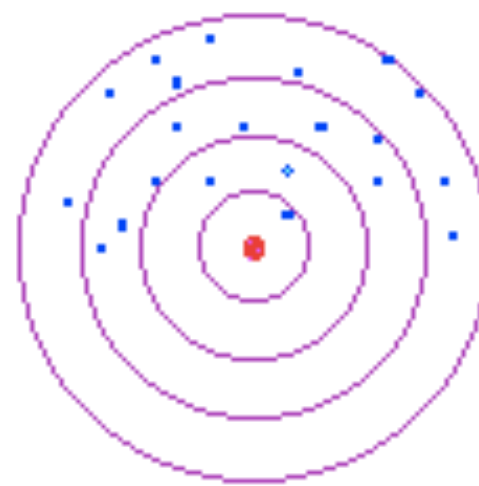
- Reliability: Does the study get consistent results?
- Validity: Does the study get true results?



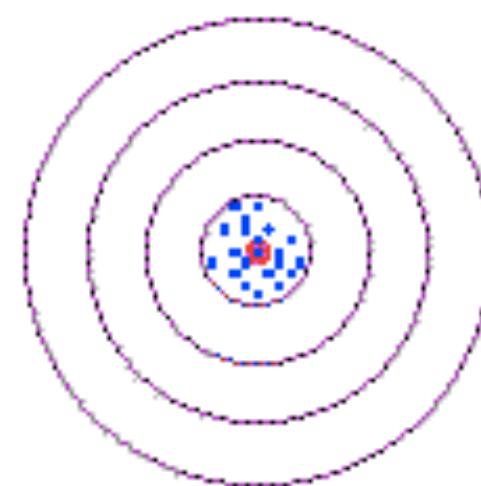
**Reliable
Not Valid**



**Valid
Not Reliable**



**Neither Reliable
Nor Valid**



**Both Reliable
And Valid**

Validity (positivist view)

○ Construct Validity

- Are we measuring the construct we intended to measure?
- Did we translate these constructs correctly into observable measures?
- Did the metrics we use have suitable discriminatory power?

○ Internal Validity

- Do the results really follow from the data?
- Have we properly eliminated any confounding variables?

○ External Validity

- Are the findings generalizable beyond the immediate study?
- Do the results support the claims of generalizability?

○ Empirical Reliability

- If the study was repeated, would we get the same results?
- Did we eliminate all researcher biases?

Typical Problems

○ Construct Validity

- Using things that are easy to measure instead of the intended concept
- Wrong scale; insufficient discriminatory power

○ Internal Validity

- Confounding variables: Familiarity and learning;
- Unmeasured variables: time to complete task, quality of result, etc.

○ External Validity

- Task representativeness: toy problem?
- Subject representativeness: students for professional developers!

○ Theoretical Reliability

- Researcher bias: subjects know what outcome you prefer



You Gotta Have A Theory

Steve Easterbrook

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www.cs.toronto.edu/~sme

Meet Stuart Dent

→ Name:

↳ Stuart Dent (a.k.a. "Stu")

→ Advisor:

↳ Prof. Helen Back

→ Topic:

↳ Merging Stakeholder views in Model Driven Development

→ Status:

↳ 2 years into his PhD

↳ Has built a tool

↳ Needs an evaluation plan





Stu's Evaluation Plan

→ Formal Experiment

- ↳ Independent Variable: Stu-Merge vs. Rational Architect
- ↳ Dependent Variables: Correctness, Speed, Subjective Assessment
- ↳ Task: Merging Class Diagrams from two different stakeholders' models
- ↳ Subjects: Grad Students in SE
- ↳ H_1 : "Stu-Merge produces correct merges more often than RA"
- ↳ H_2 : "Subjects produce merges faster with Stu-Merge than with RA"
- ↳ H_3 : "Subjects prefer using Stu-Merge to RA"

→ Results

- ↳ H_1 accepted (strong evidence)
- ↳ H_2 & H_3 rejected
- ↳ Subjects found the tool unintuitive



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What could have gone wrong?



Threats to Validity

→ Construct Validity

- ↳ What do we mean by a merge? What is correctness?
- ↳ 5-point scale for subjective assessment - insufficient discriminatory power
 - (both tools scored very low)

→ Internal Validity

- ↳ Confounding variables: Time taken to learn the tool; familiarity
- ↳ Subjects were all familiar with RA, not with Stu-merge

→ External Validity

- ↳ Task representativeness: class models were of a toy problem
- ↳ Subject representativeness: Grad students as sample of what population?

→ Theoretical Reliability

- ↳ Researcher bias: subjects knew Stu-merge was Stu's own tool

More on validity in the backup slides at the end of the talk



What went wrong?

- What was the research question?
 - ↳ "Is tool A better than tool B?"
- What would count as an answer?
- What use would the answer be?
 - ↳ How is it a "contribution to knowledge"?
- How does this evaluation relate to the existing literature?

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Stu's Theory

→ Background Assumptions

- ↳ Large team projects, models contributed by many actors
- ↳ Models are fragmentary, capture partial views
- ↳ Partial views are inconsistent and incomplete most of the time

→ Basic Theory

- ↳ (Brief summary:)
- ↳ Model merging is an exploratory process, in which the aim is to discover intended relationships between views. 'Goodness' of a merge is a subjective judgment. If an attempted merge doesn't seem 'good', many need to change either of the models, or the way in which they were mapped together.

→ Derived Hypotheses

- ↳ Useful merge tools need to represent relationships explicitly
- ↳ Useful merge tools need to be complete (work for any models, even if inconsistent)

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Stu's Research Question(s)

→ Existence

↳ Does model merging ever happen in practice?

→ Description/Classification

↳ What are the different types of model merging that occur in practice on large scale systems?

→ Descriptive-Comparative

↳ How does model merging with explicit representation of relationships differ from model merging without such representation?

→ Causality

↳ Does an explicit representation of the relationship between models cause developers to explore different ways of merging models?

→ Causality-Comparative

↳ Does the algebraic representation of relationships in Stu's tool lead developers to explore more than do pointcuts in AOM?

Picture just one for now...



Stu's Method(s) Selection...

→ Existence

↳ Does model merging ever happen in practice?

→ Description/Classification

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→ Descriptive-Comparative

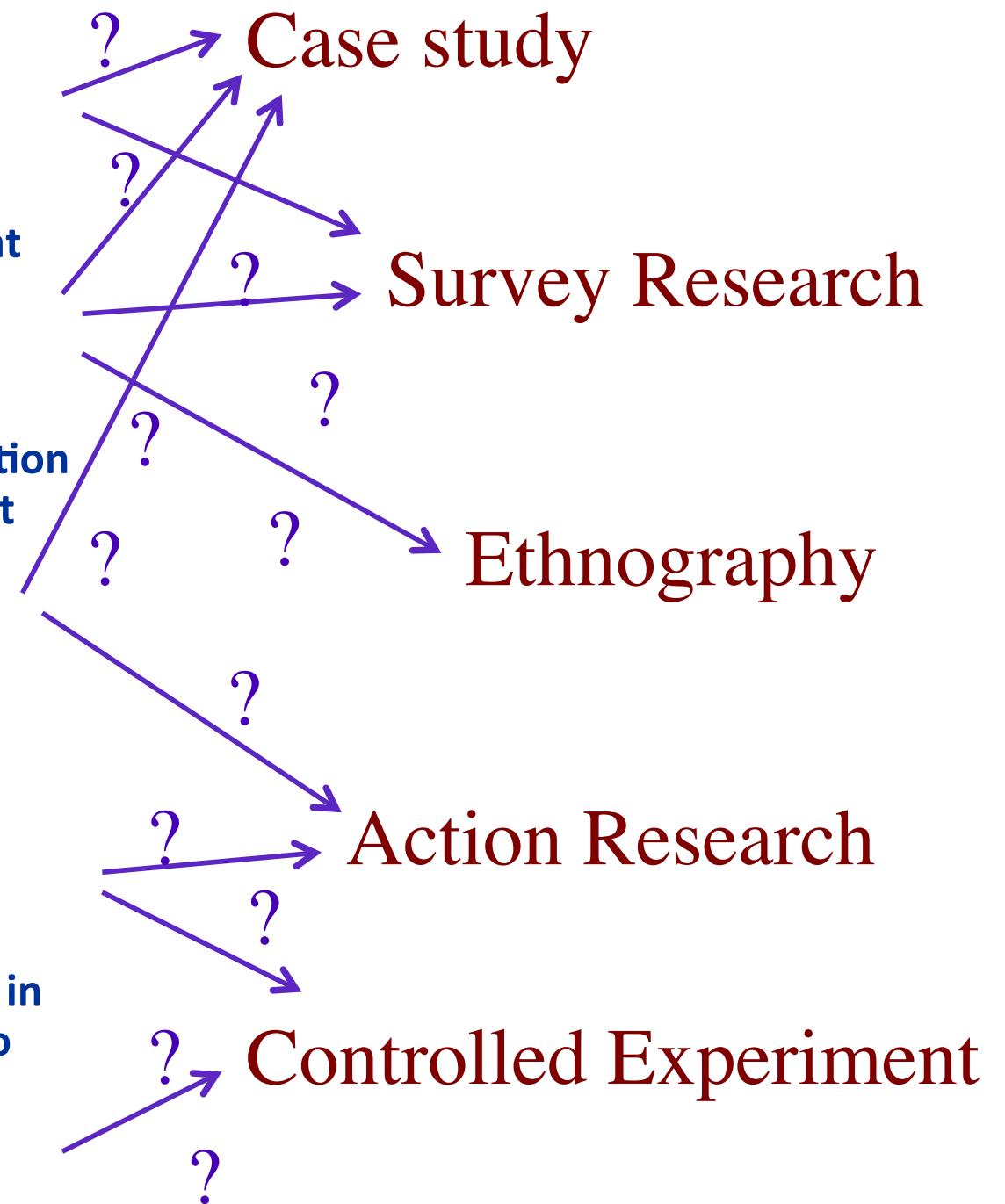
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↳ Does the algebraic representation of relationships in Stu's tool lead developers to explore more than do pointcuts in AOM?





Warning

No method is perfect

Don't get hung up on methodological purity

Pick something and get on with it

Some knowledge is better than none



Why Build a Tool?

→ Build a Tool to Test a Theory

↳ Tool is part of the experimental materials needed to conduct your study

→ Build a Tool to Develop a Theory

↳ Theory emerges as you explore the tool

→ Build a Tool to Explain your Theory

↳ Theory as a concrete instantiation of (some aspect of) the theory

Why did we
build tools?



Why do you
build tools?